Duration Models and Treatment Effects

Motivated by Abbring & Van den Berg (2003, *Econometrica*)

Econometrics Reading Group

Why duration models?

- Many economic outcomes are about **how long until something happens**:
  - unemployment → job; poverty → exit; firm → closure; marriage → divorce.
- The questions are about the **rate** of leaving:
  - how it changes with time already spent;
  - how it responds to events along the way.

Destination of this talk: Abbring & Van den Berg (2003)

- The effect of a treatment that *arrives during the spell* (e.g. a benefits reduction) on the exit rate.
- To get there, we build the objects one at a time.

Road map: the questions, in order

1. **How do we describe a duration?**
   Hazard and survival. Cox (1972), Lancaster (1979)

2. **People differ: is a falling exit rate duration dependence or sorting?**
   Unobserved heterogeneity, dynamic selection. Lancaster (1979), Heckman & Singer (1984)

3. **When can the two be separated?**
   The mixed proportional hazard model. Elbers & Ridder (1982)

4. **Two durations.**
   Competing risks. Cox (1959), Tsiatis (1975), Heckman & Honoré (1989)

5. **A treatment arriving during the spell.**
   Abbring & Van den Berg (2003)

One duration: hazard and survival

$Y \geq 0$ continuous, $S(y) := \Pr(Y > y)$ survival, f density.

$$\theta(y) := \lim_{dy \downarrow 0} \frac{\Pr(y \leq Y < y + dy \mid Y \geq y)}{dy} = \frac{f(y)}{S(y)} \qquad \text{[conditional probability]}$$

$$\Theta(y) := \int_0^y \theta(u)\, du = \int_0^y -\frac{S'(u)}{S(u)}\, du = -\log S(y) \qquad [S' = -f;\ S(0) = 1]$$

$$\implies \quad S(y) = e^{-\Theta(y)}$$

- $\theta(y)$: exit rate *among those still unemployed*. The effect will act on θ.
- **Duration dependence**: how $\theta(y)$ changes with time already spent.
  - θ increasing: *positive* (the longer unemployed, the more likely to leave).
  - θ decreasing: *negative* (the longer unemployed, the less likely to leave).
  - θ constant: none. $S(y) = e^{-cy}$, memoryless.

▸ Proof

People differ: individual vs. observed hazard

Unobserved type $V \geq 0$, $V \sim G$. Assume

$$\theta(y \mid V) = \lambda(y)\, V, \qquad \lambda(y) = \theta(y \mid V = 1), \qquad S(y \mid V) = e^{-\Lambda(y)V}.$$

Data reveal the **observed** hazard, averaged over types:

$$\bar{S}(y) = \mathbb{E}\big[S(y \mid V)\big] = \mathbb{E}\big[e^{-\Lambda(y)V}\big] \qquad \text{[LIE]}$$

$$\bar{\theta}(y) = \frac{\bar{f}(y)}{\bar{S}(y)} = -\frac{\bar{S}'(y)}{\bar{S}(y)} = \lambda(y)\, \frac{\mathbb{E}\big[V e^{-\Lambda(y)V}\big]}{\mathbb{E}\big[e^{-\Lambda(y)V}\big]} \qquad \text{[differentiate under } \mathbb{E}\text{]}$$

- The ratio is an average of V, weighting each type by its survival $e^{-\Lambda(y)V}$.
- Next: it is the average type **among those still unemployed**.

▸ Proof

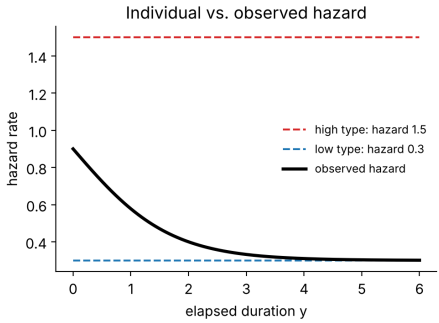
Who is still unemployed?

$$\Pr(V = v \mid Y > y) = \frac{S(y \mid v)\Pr(V = v)}{\bar{S}(y)} \qquad \text{[Bayes]}$$

$$\mathbb{E}[V \mid Y > y] = \frac{\mathbb{E}[Ve^{-\Lambda(y)V}]}{\mathbb{E}[e^{-\Lambda(y)V}]} \qquad \text{[its mean]}$$

$$\implies \bar{\theta}(y) = \lambda(y)\,\mathbb{E}[V \mid Y > y] \qquad \text{[previous slide]}$$

- Prior $\Pr(V = v) \times$ likelihood $S(y \mid v)$: staying is evidence of a low v.
- Average individual hazard: $\lambda(y)\mathbb{E}[V]$.
- Observed hazard: $\lambda(y)\mathbb{E}[V \mid Y > y]$.
- Gap = **dynamic selection**.



Two types, constant hazards. Nobody's hazard falls; the observed one does.

▸ Proof

The Laplace transform of G

For a distribution G of $V \geq 0$, define the **function**

$$\mathcal{L}_G : [0,\infty) \to (0,1], \qquad \mathcal{L}_G(a) := \mathbb{E}_G\left[e^{-aV}\right] = \int_0^\infty e^{-av}\, dG(v).$$

- *Which* function $\mathcal{L}_G$ is depends only on G:
  - $V \equiv 1$: $\mathcal{L}_G(a) = e^{-a}$; two types v_1, v_2 w.p. $p, 1-p$: $\mathcal{L}_G(a) = p\, e^{-av_1} + (1-p)\, e^{-av_2}$.

Link to the data: plug in $a = \Lambda(y)$.

$$\bar{S}(y) = \mathbb{E}_G\left[e^{-\Lambda(y)V}\right] = \mathcal{L}_G\big(\Lambda(y)\big) \qquad \text{[definition, } a = \Lambda(y)\text{]}$$

$$\bar{\theta}(y) = \underbrace{\lambda(y)}_{\text{from } \Lambda} \cdot \underbrace{m_G\big(\Lambda(y)\big)}_{\text{from } G}, \qquad m_G(a) := -\frac{\mathcal{L}_G'(a)}{\mathcal{L}_G(a)} \qquad \text{[}\bar{\theta} = -\tfrac{d}{dy}\log\bar{S}\text{; chain rule]}$$

- $\bar{S} = \mathcal{L}_G \circ \Lambda$: first $y \mapsto \Lambda(y)$ (duration dependence), then $a \mapsto \mathcal{L}_G(a)$ (heterogeneity).
- $m_G(a) = \mathbb{E}[V \mid \text{survived to } a]$ and $m_G'(a) = -\operatorname{Var}(V \mid \text{survived to } a) \leq 0$: sorting.

▸ Proof

What $\mathcal{L}_G$ pins down

Uniqueness

$$\mathcal{L}_{G_1}(a) = \mathcal{L}_{G_2}(a) \;\; \forall a \geq 0 \quad \Longleftrightarrow \quad G_1 = G_2.$$

- $G_1 \neq G_2 \Rightarrow$ different $\bar{S} = \mathcal{L}_G(\Lambda(y))$ as functions of $\Lambda(y)$.

Analyticity: for any $0 < a_{lo} < a_{hi}$,

$$\mathcal{L}_{G_1}(a) = \mathcal{L}_{G_2}(a) \;\; \forall a \in [a_{lo}, a_{hi}] \quad \Longrightarrow \quad \mathcal{L}_{G_1}(a) = \mathcal{L}_{G_2}(a) \;\; \forall a > 0.$$

- Combined with uniqueness: $G_1 \neq G_2 \Rightarrow \mathcal{L}_{G_1} \neq \mathcal{L}_{G_2}$ on *every* interval.
- They cannot match on a stretch and then diverge, so knowing $\mathcal{L}_G$ on a stretch is knowing it everywhere.

So identifying G reduces to one task: observe $\mathcal{L}_G(a)$ on some interval of a, i.e. the observed survival $\bar{S}(y)$ at *known* values of the accumulated baseline hazard $a = \Lambda(y)$.

▸ Proof

What we lack: the baseline Λ

What the data give

- Observed: $\bar{S}(y)$, equivalently $\bar{\theta}(y)$, for all y.
- Not observed: $\Lambda(y) = \int_0^y \lambda$, the baseline of the *individual* hazard.
- In data it is mixed with sorting: $\bar{\theta}(y) = \lambda(y)\, m_G(\Lambda(y))$.

If Λ were known

$$\mathcal{L}_G(a) = \bar{S}\big(\Lambda^{-1}(a)\big) \quad \text{for } a \text{ in the range of } \Lambda \qquad [\bar{S}(y) = \mathcal{L}_G(\Lambda(y)), \text{ set } y = \Lambda^{-1}(a)]$$

- $\Rightarrow \mathcal{L}_G$ on an interval $\Rightarrow$ (analyticity) everywhere $\Rightarrow$ (uniqueness) G.
- E.g. $\Lambda(y) = y$: then $\mathcal{L}_G = \bar{S}$. The observed survival curve *is* the Laplace transform of G.

But Λ is unknown: we see the values $\mathcal{L}_G(\Lambda(y))$, not the points $\Lambda(y)$ where they sit.

$\rightarrow$ Can $\bar{S}$ alone separate Λ from G?

Duration Dependence or Sorting? Not Identified

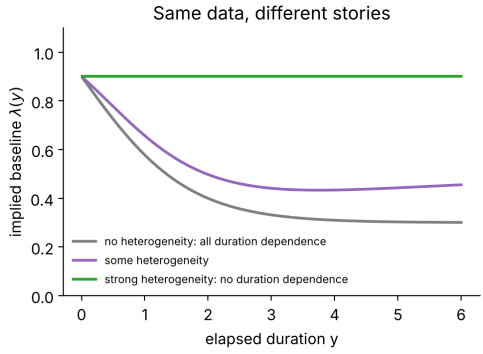
- Observed: $\bar{S}(y)$, equivalently $\bar{\theta}(y)$.
- Not observed: λ (duration dependence) and G (sorting).
- They are mixed in one observed function of y:

$$\bar{\theta}(y) = \underbrace{\lambda(y)}_{\text{duration dep.}} \cdot \underbrace{\mathbb{E}[V \mid Y > y]}_{\text{sorting}}$$

- Any sorting pattern can be offset by a matching λ: every G fits the data (figure).

To separate them, we need variation that moves one but not the other:

- groups at different stages of sorting, same λ;
- the same person observed twice;
- parametric restrictions on Λ or G.



Same observed survival, three type distributions, three implied baselines.

Proof

Covariates separate sorting from duration dependence

MPH model: $\theta(y \mid x, V) = \lambda(y)\,\varphi(x)\,V, \quad V \perp X$. Data: $\bar{S}(y \mid x)$ for all (y, x).

$$\bar{S}(y \mid x) = \mathcal{L}\big(\varphi(x)\Lambda(y)\big): \quad \text{one } \mathcal{L} \text{ and one } \Lambda \text{ must fit } \textit{all} \text{ groups.}$$

Let $m(a) := -\mathcal{L}'(a)/\mathcal{L}(a) = \mathbb{E}[V \mid \text{survived to } a]$, decreasing. Compare $\varphi_1 > \varphi_2$:

$$\frac{\bar{\theta}(y \mid x_1)}{\bar{\theta}(y \mid x_2)} = \frac{\varphi_1}{\varphi_2} \cdot \frac{m(\varphi_1\Lambda(y))}{m(\varphi_2\Lambda(y))} \qquad [\lambda(y) \text{ cancels}]$$

$$< \frac{\varphi_1}{\varphi_2} \;\; (y > 0), \qquad = \frac{\varphi_1}{\varphi_2} \;\; (y = 0) \qquad [\varphi_1\Lambda > \varphi_2\Lambda,\ m \text{ decreasing}]$$

- At the same duration, the high-φ group is further along: its sorting is more advanced.
- Duration dependence hits both groups equally and cancels. Sorting does not.
- So the **drift of the ratio below φ_1/φ_2 measures sorting**.
- MPH reads: *absent sorting, covariate effects would be constant over duration*.

▸ Proof

MPH identification in three steps

Elbers & Ridder (1982)

Normalize $\varphi(x^*) = 1$, $\Lambda(y^*) = 1$. Assume $\mathbb{E}V < \infty$ and variation in $\varphi(x)$.

1. $\dfrac{\bar{\theta}(y \mid x)}{\bar{\theta}(y \mid x^*)} \xrightarrow{y \downarrow 0} \varphi(x)$ [nobody sorted yet: $m(0)/m(0) = 1$]

2. $\bar{S}(y^* \mid x) = \mathcal{L}\big(\varphi(x)\big)$ [$\Lambda(y^*) = 1$]

3. $\Lambda(y) = \mathcal{L}^{-1}\big(\bar{S}(y \mid x)\big)/\varphi(x)$ [$\mathcal{L}$ strictly decreasing]

- Step 2: varying x traces $\mathcal{L}$ on an interval $\Rightarrow$ (analyticity) everywhere $\Rightarrow$ (uniqueness) G.
- One duration with covariates reveals λ, φ and G.
- Price: proportionality, $V \perp X$, $\mathbb{E}V < \infty$; identification leans on short durations.

Proof

Adding a Treatment

A second duration: the treatment

Where it comes from

- A treatment that *arrives* at a random time S, with its own hazard, is itself a duration.
- Before S: **competing risks** between Y (job) and S (treatment); whichever comes first.
- After S: Y continues under treatment. The effect shows up here.

Why it is a hurdle

- Censoring: if the job comes first, S is never seen.
- Endogeneity: the same unobserved traits may drive both.
- Tsiatis (1975): competing-risks data alone cannot reveal that dependence.

Why modelling it is the solution

- If S were exogenous, it would be an ordinary time-varying covariate.
- Otherwise, model S's hazard: a dynamic **selection equation**.
  - Before S there is no effect yet: a clean window to learn the selection.

Competing risks: model and data

$$\theta_Y(t \mid x, V) = \lambda_Y(t)\varphi_Y(x)V_Y, \qquad \theta_S(t \mid x, V) = \lambda_S(t)\varphi_S(x)V_S, \qquad (V_Y, V_S) \sim G.$$

Assumption: $Y \perp S \mid (x, V)$. All dependence runs through (x, V).

$$\begin{aligned} \Pr(Y > y, S > s \mid x) &= \mathbb{E}\big[e^{-\varphi_Y \Lambda_Y(y) V_Y} e^{-\varphi_S \Lambda_S(s) V_S}\big] && \text{[LIE; assumption]} \\ &= \mathcal{L}\Big(\underbrace{\varphi_Y(x)\Lambda_Y(y)}_{=:\Lambda_Y(y|x)}, \underbrace{\varphi_S(x)\Lambda_S(s)}_{=:\Lambda_S(s|x)}\Big) && [\mathcal{L}(a,b) := \mathbb{E}[e^{-aV_Y - bV_S}]] \end{aligned}$$

Data: only the first event and which one, i.e. the diagonal $y = s = t$:

$$\begin{aligned} \bar{S}_0(t \mid x) &:= \Pr(Y > t,\, S > t \mid x) = \mathcal{L}\big(\Lambda_Y(t \mid x), \Lambda_S(t \mid x)\big) && \text{[nothing happened by } t\text{]} \\ \bar{\theta}_0^Y(t \mid x) &:= \lim_{dt \downarrow 0} \frac{\Pr(t \le Y < t + dt \mid Y \ge t,\, S \ge t,\, x)}{dt} && \text{[exit rate among the risk set]} \\ &= \lambda_Y(t)\varphi_Y(x)\,\mathbb{E}[V_Y \mid R_t, x] && \text{[see proof]} \end{aligned}$$

- $R_t := \{Y \ge t,\, S \ge t\}$: still unemployed and untreated. Subscript 0: before treatment; superscript: which exit. $\bar{\theta}_0^Y + \bar{\theta}_0^S = \bar{\theta}_0$, the observed hazard of leaving by either exit.

▸ Proof

The cause-specific hazard is filtered by the other exit

Observed: job-finding rate among those still unemployed *and* untreated

$$\bar{\theta}_0^Y(t \mid x) = \lambda_Y(t)\varphi_Y(x)\,\mathbb{E}[V_Y \mid Y \geq t,\ \mathbf{S} \geq \mathbf{t},\ x]$$

Not observed: the one-duration observed hazard of Y, now cut short by treatment

$$\bar{\theta}_Y(t \mid x) = \lambda_Y(t)\varphi_Y(x)\,\mathbb{E}[V_Y \mid Y \geq t,\ x]$$

What changes: with $W := e^{-\Lambda_S(t|x)V_S} = \Pr(\text{still untreated} \mid V_S)$, all conditional on $Y \geq t, x$:

$$\begin{aligned}
\mathbb{E}[V_Y \mid S \geq t] &= \frac{\mathbb{E}[V_Y\, W]}{\mathbb{E}[W]} && \text{[Bayes: reweight by } W\text{]} \\
&= \mathbb{E}[V_Y] + \frac{\mathrm{Cov}(V_Y, W)}{\mathbb{E}[W]} && \text{[}\mathbb{E}XY = \mathbb{E}X\,\mathbb{E}Y + \mathrm{Cov}(X,Y)\text{]}
\end{aligned}$$

- $V_S \perp V_Y$: $\mathrm{Cov} = 0$, so $\bar{\theta}_0^Y = \bar{\theta}_Y$. Treatment removes people at random in V_Y.
- Correlated: $\mathrm{Cov} \neq 0$. W falls in V_S, so the sign is opposite to the correlation of V_Y, V_S.
  - Low motivation: low V_Y, high V_S $\Rightarrow$ the untreated are left with *higher* V_Y.

Why selection matters for the treatment effect

- The effect is measured by comparing, at the same s, the **just treated** with the **not yet treated**.
- Who is treated at s: those at risk, reweighted by V_S.
  - V_S correlated with $V_Y \Rightarrow$ the treated have a different V_Y mix.
- Then

$$\frac{\text{exit rate, just treated}}{\text{exit rate, not yet treated}} = \delta \times (\text{selection factor}).$$

- A higher exit rate after treatment: does treatment work, or are the treated fast types anyway?
  - Without the dependence of V_Y, V_S we cannot tell. The duration analogue of an endogenous regressor.
- **So:** identify the dependence *before* treatment, where no effect is present.

$\rightarrow$ Where does it show up in the data?

Where the selection lives

Selection = dependence of V_Y and V_S among those at risk:

$$\frac{\partial^2 \log \bar{S}_0}{\partial a \, \partial b} = \mathrm{Cov}(V_Y, V_S \mid R_t, x) \qquad \text{[quotient rule]}$$

$$\frac{\partial \log \bar{\theta}_0^Y}{\partial b} = -\frac{\mathrm{Cov}(V_Y, V_S \mid R_t, x)}{\mathbb{E}[V_Y \mid R_t, x]} \qquad \text{[hold } a \text{ fixed]}$$

- Raise treatment intensity, hold the exit index fixed. Those at risk are untreated, so no causal channel.
- Any response of their exit rate is **pure selection**.

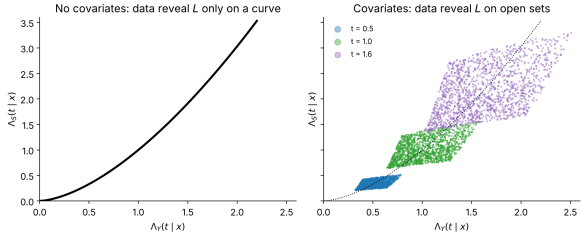
Assumption 3: $\{(\varphi_Y(x), \varphi_S(x))\}$ contains an open set of $\mathbb{R}^2$.

- At least two continuous covariates that cannot affect both hazards in the same proportions. One covariate only traces a curve.
- E.g. $\varphi_j = e^{x'\beta_j}$, two covariates: direction $d = (\beta_{Y2}, -\beta_{Y1})$ gives $d'\beta_Y = 0$, $d'\beta_S \neq 0$ iff $(\beta_Y \ \beta_S)$ full rank.
- **vs. exclusion restriction:** no variable is excluded; a *direction* is, and it is learned from data.

▸ Proof

Tsiatis' problem and how covariates solve it

- The surface $\mathcal{L}(a, b) = \Pr(\text{nothing yet})$ at indices (a, b). Selection lives in its shape.
- Each (group x, duration t) gives its height $\bar{S}_0(t \mid x)$ at one location.
- **No covariates:** locations lie on one curve $t \mapsto (\Lambda_Y(t), \Lambda_S(t))$. Cannot pin down a surface.
- **Covariates (Assn. 3):** locations fill a 2-D patch $\Rightarrow$ surface $\Rightarrow$ G.



Three steps, as before:

1. $\bar{\theta}_0^Y, \bar{\theta}_0^S$ ratios as $t \downarrow 0 \Rightarrow \varphi_Y, \varphi_S$
2. $\bar{S}_0(t^* \mid x) = \mathcal{L}(\varphi_Y(x), \varphi_S(x)) \Rightarrow G$
3. solve for $\Lambda_Y(t), \Lambda_S(t)$

▸ Proof

Adding the treatment effect

Abbring & Van den Berg (2003)

- Potential outcomes indexed by treatment date: $Y^*(s)$, $s \in [0, \infty]$. Observed $Y = Y^*(S)$.
- **No anticipation:** $\theta_{Y^*(s)}(t) = \theta_{Y^*(\infty)}(t)$ for $t < s$. The future date does not affect today's hazard.
- **Selection on** (x, V)**:** $Y^*(\cdot) \perp S \mid (x, V)$.

Model 1a

$$\theta_Y(t \mid s, x, V) = \lambda_Y(t)\varphi_Y(x)V_Y \cdot \delta(t \mid s, x)^{\mathbf{1}\{t>s\}}, \qquad \theta_S(t \mid x, V) = \lambda_S(t)\varphi_S(x)V_S.$$

- δ: the causal effect, a multiplier on the exit rate after treatment. May vary with t, s, x.
- Before S: exactly the competing-risks model. So φ's, Λ's and G are identified.
- After S: the treated cohort's survival $\bar{S}_1(y \mid s) = \Pr(Y > y \mid S = s, Y > s)$ carries δ.
- Given (x, V), the timing of S is still pure luck: the instrument-like variation.

▸ Proof

Who is treated at s? The selection factor

$$dG_{R_s}(v) \propto e^{-a_s v_Y - b_s v_S}\, dG(v) \qquad \text{[Bayes: survivors at } s\text{]}$$

$$dG_s(v) \propto v_S\, dG_{R_s}(v) \qquad \text{[treated in } [s, s+ds) \text{ w.p. } \propto v_S\text{]}$$

Exit rate right after s, treated vs. not-yet-treated:

$$\frac{\text{just treated}}{\text{not yet treated}} = \delta(s \mid s, x) \cdot \frac{\mathbb{E}_{G_s}[V_Y]}{\mathbb{E}_{R_s}[V_Y]} \qquad [\lambda_Y \varphi_Y \text{ cancel}]$$

$$= \delta(s \mid s, x) \cdot \left(1 + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s} V_Y \; \mathbb{E}_{R_s} V_S}\right) \qquad [\text{reweight by } v_S]$$

- Observed ratio = causal effect $\times$ **selection factor**.
- No dependence $\Rightarrow$ factor 1: timing as good as random.
- The factor is a function of $\mathcal{L}$, known from the competing-risks part $\Rightarrow$ δ identified.

▸ Proof

After the moment of treatment

With $D(y \mid s,x) := \int_s^y \lambda_Y(\tau)\delta(\tau \mid s,x)\,d\tau$:

$$\bar{S}_1(y \mid s,x) = \mathbb{E}_{G_s}\left[e^{-\varphi_Y(x)V_Y D(y|s,x)}\right] \qquad \text{[LIE over the treated cohort]}$$

$$= \frac{\mathcal{L}_b\big(\Lambda_Y(s \mid x) + \varphi_Y(x)D,\ \Lambda_S(s \mid x)\big)}{\mathcal{L}_b(\Lambda_Y(s \mid x), \Lambda_S(s \mid x))} \qquad \text{[the paper's eq. (10)]}$$

- Left: data. Right: known except D. Decreasing in $D \Rightarrow$ invert $\Rightarrow \delta = \partial_y D/\lambda_Y$.
- Two selection problems, both handled by knowing G:
  1. **into** treatment at s: the v_S tilt;
  2. **out of** the treated cohort after s: its survivors drift to low V_Y, even with random timing.

pre-treatment competing risks $\Rightarrow \varphi, \Lambda, G \Rightarrow$ who is treated at $s \Rightarrow + \bar{S}_1 \Rightarrow \delta$

▸ Proof

What is traded

Not needed

- an excluded instrument
- selection on observables only
- parametric forms

Needed

- no anticipation (untestable)
- time-invariant unobservables, $V \perp X$
- proportional hazards (MPH)
- non-proportional covariate variation
- finite means; continuous time

The idea in one line

- The treatment acts only *after* S; unobserved types act throughout.
- The excluded *variable* becomes an excluded timing *shock*.

Caveats

- Identification leans on short durations.
- Discrete data blur the timing.
- δ is a hazard ratio given V, not a population effect.

Connections

Staggered-adoption DiD

- $Y^*(s) \leftrightarrow Y_t(g)$; the same no-anticipation assumption; risk set $\leftrightarrow$ not-yet-treated.
- Difference: an absorbing exit changes who remains, so parallel trends in hazards fails even with random timing.

Sequential exogeneity and feedback

- A hazard is an intensity given the past: a continuous-time sequential moment condition.
- Single spell: treatment may depend on permanent V, not on transitory shocks.

State dependence vs. permanent heterogeneity

- λ (duration dependence), G (heterogeneity), δ (intervention): the framework separates all three.
- Multiple spells identify more, but assume no causal links across spells.

Proof: hazard and survival

$$\theta(y) = \lim_{dy \downarrow 0} \frac{\Pr(y \le Y < y + dy \mid Y \ge y)}{dy} \qquad \text{[definition]}$$

$$= \lim_{dy \downarrow 0} \frac{\Pr(y \le Y < y + dy)}{dy \, \Pr(Y \ge y)} \qquad [\Pr(A \mid B) = \Pr(A \cap B)/\Pr(B)]$$

$$= \lim_{dy \downarrow 0} \frac{F(y + dy) - F(y)}{dy} \cdot \frac{1}{S(y)} = \frac{f(y)}{S(y)} \qquad [Y \text{ continuous; } F' = f]$$

$$\Theta(y) = \int_0^y \frac{f(u)}{S(u)} du = \int_0^y -\frac{S'(u)}{S(u)} du \qquad [S = 1 - F \Rightarrow S' = -f]$$

$$= -\big[\log S(u)\big]_0^y = -\log S(y) \qquad [(\log S)' = S'/S;\ S(0) = 1]$$

$$\implies S(y) = e^{-\Theta(y)}, \qquad f(y) = \theta(y) e^{-\Theta(y)} \qquad \text{[exponentiate; } f = \theta S]$$

Constant hazard c: $\Theta = cy$, and $\Pr(Y > y + u \mid Y > y) = S(y + u)/S(y) = e^{-cu}$, free of y.

Back

Proof: observed survival and hazard

Model: $\theta(y \mid V) = \lambda(y)V$, so $\Theta(y \mid V) = \Lambda(y)V$ and $S(y \mid V) = e^{-\Lambda(y)V}$.

$$\begin{aligned}
\bar{S}(y) &= \Pr(Y > y) = \mathbb{E}\big[\Pr(Y > y \mid V)\big] && \text{[LIE]} \\
&= \mathbb{E}\big[e^{-\Lambda(y)V}\big] && [S = e^{-\Theta} \text{ for each type}] \\
\bar{f}(y) &= -\bar{S}'(y) = \mathbb{E}\big[\lambda(y)Ve^{-\Lambda(y)V}\big] && \text{[differentiate under } \mathbb{E};\ \Lambda' = \lambda] \\
\bar{\theta}(y) &= \frac{\bar{f}(y)}{\bar{S}(y)} = \lambda(y)\,\frac{\mathbb{E}\big[Ve^{-\Lambda(y)V}\big]}{\mathbb{E}\big[e^{-\Lambda(y)V}\big]} && [\lambda(y) \text{ not random}]
\end{aligned}$$

- $\lambda(y) = \theta(y \mid V = 1)$ needs a normalization: (λ, V) and $(c\lambda, V/c)$ give the same hazard.

Back

Proof: types among survivors

$$dG(v \mid Y > y) = \frac{\Pr(Y > y \mid V = v)\, dG(v)}{\Pr(Y > y)} \qquad \text{[Bayes]}$$

$$= \frac{e^{-\Lambda(y)v}\, dG(v)}{\mathbb{E}\left[e^{-\Lambda(y)V}\right]} \qquad [S(y \mid v),\ \bar{S}(y)]$$

$$\mathbb{E}[V \mid Y > y] = \int v\, dG(v \mid Y > y) \qquad \text{[conditional mean]}$$

$$= \frac{\int v\, e^{-\Lambda(y)v}\, dG(v)}{\mathbb{E}\left[e^{-\Lambda(y)V}\right]} \qquad \text{[plug in; denominator constant in } v\text{]}$$

$$= \frac{\mathbb{E}\left[Ve^{-\Lambda(y)V}\right]}{\mathbb{E}\left[e^{-\Lambda(y)V}\right]} \qquad [\textstyle\int h\, dG = \mathbb{E}[h(V)]]$$

$$\implies \bar{\theta}(y) = \lambda(y)\, \mathbb{E}[V \mid Y > y] \qquad \text{[same ratio as in } \bar{\theta}\text{]}$$

Back

Proof: Laplace transform properties

$\mathcal{L}(a) = \mathbb{E}[e^{-aV}]$; let $\mathbb{E}_w$ denote expectation under weights $\propto e^{-aV}$ (survivors at index a).

$$\mathcal{L}'(a) = -\mathbb{E}\big[Ve^{-aV}\big], \qquad \mathcal{L}''(a) = \mathbb{E}\big[V^2 e^{-aV}\big] \qquad \text{[differentiate under } \mathbb{E}\text{]}$$

$$-\frac{\mathcal{L}'}{\mathcal{L}} = \mathbb{E}_w[V] = \mathbb{E}[V \mid \text{survived to } a] \qquad \text{[as in the Bayes proof, } \Lambda = a\text{]}$$

$$(\log \mathcal{L})'' = \frac{\mathcal{L}''}{\mathcal{L}} - \Big(\frac{\mathcal{L}'}{\mathcal{L}}\Big)^2 = \mathbb{E}_w[V^2] - \mathbb{E}_w[V]^2 = \mathrm{Var}_w(V) \qquad \text{[quotient rule]}$$

$$\frac{d}{dy}\mathbb{E}[V \mid Y > y] = \frac{d}{dy}\Big[-(\log \mathcal{L})'\big(\Lambda(y)\big)\Big] \qquad \text{[}\mathbb{E}[V \mid Y > y] = -(\log \mathcal{L})'(\Lambda(y))\text{]}$$

$$= -(\log \mathcal{L})''\big(\Lambda(y)\big)\,\lambda(y) = -\lambda(y)\,\mathrm{Var}(V \mid Y > y) \qquad \text{[chain rule]}$$

▸ Uniqueness and analyticity

◂ Back

Proof: uniqueness and analyticity

Uniqueness. A distribution on $[0,\infty)$ is determined by its Laplace transform (standard; e.g. Feller, Vol. II, XIII.1).

- Intuition: $(-1)^k \mathcal{L}^{(k)}(0) = \mathbb{E}[V^k]$; large a isolates the lowest types.

Analyticity. Fix $a_0 > 0$ and $|h| < a_0$:

$$\begin{aligned}
\mathcal{L}(a_0 + h) &= \mathbb{E}\Big[e^{-a_0 V} \sum_{k\geq 0} \frac{(-hV)^k}{k!}\Big] && \text{[Taylor series of } e^{-hV}\text{]} \\
&= \sum_{k\geq 0} \frac{(-h)^k}{k!} \mathbb{E}\big[V^k e^{-a_0 V}\big] && \text{[Fubini: } \mathbb{E}[e^{-(a_0-|h|)V}] < \infty\text{]} \\
&= \sum_{k\geq 0} \frac{\mathcal{L}^{(k)}(a_0)}{k!}\, h^k && [\mathcal{L}^{(k)}(a_0) = (-1)^k \mathbb{E}[V^k e^{-a_0 V}]]
\end{aligned}$$

- If $\mathcal{L}$ is known on an interval, all derivatives at an interior point are known, so $\mathcal{L}$ is known on a neighbourhood. Chain neighbourhoods across $(0,\infty)$.
- Same in two dimensions, but an *open set* of $\mathbb{R}^2$ is needed: a curve is not enough.

Back

Proof: any G fits one survival curve

Data $\bar{S}$, continuous and strictly decreasing from 1 to 0. Pick any $\tilde{G}$ with $\Pr(\tilde{V} = 0) = 0$. We need $\mathcal{L}_{\tilde{G}}(\tilde{\Lambda}(y)) = \bar{S}(y)$ for all y.

$$\mathcal{L}_{\tilde{G}} : [0, \infty) \to (0, 1] \text{ strictly decreasing, onto} \qquad [\mathcal{L}(0) = 1,\ \mathcal{L}' < 0,\ \mathcal{L}(\infty) = \Pr(\tilde{V} = 0) = 0]$$

$$\tilde{\Lambda}(y) := \mathcal{L}_{\tilde{G}}^{-1}\big(\bar{S}(y)\big) \qquad [\text{one equation, one unknown, each } y]$$

$$\tilde{\Lambda}(0) = \mathcal{L}_{\tilde{G}}^{-1}(1) = 0, \quad \tilde{\Lambda} \text{ increasing} \qquad [\bar{S}(0) = 1;\ \text{composition of decreasing maps}]$$

$$\bar{\theta}(y) = -\frac{d}{dy} \log \mathcal{L}_{\tilde{G}}\big(\tilde{\Lambda}(y)\big) = \tilde{\lambda}(y)\, \tilde{\mathbb{E}}[V \mid Y > y] \qquad [\text{chain rule; survivor mean}]$$

- So $\tilde{\lambda} = \bar{\theta} / \tilde{\mathbb{E}}[V \mid Y > y]$: whatever sorting $\tilde{G}$ implies, $\tilde{\lambda}$ absorbs the rest.

Back

Proof: covariate effects drift under heterogeneity

Let $m(a) := -\mathcal{L}'(a)/\mathcal{L}(a) = \mathbb{E}[V \mid \text{survived to } a]$.

$$\bar{\theta}(y \mid x) = -\partial_y \log \mathcal{L}\big(\varphi(x)\Lambda(y)\big) = \lambda(y)\varphi(x)\, m\big(\varphi(x)\Lambda(y)\big) \qquad \text{[chain rule]}$$

$$\frac{\bar{\theta}(y \mid x_1)}{\bar{\theta}(y \mid x_2)} = \frac{\varphi_1}{\varphi_2} \cdot \frac{m(\varphi_1\Lambda(y))}{m(\varphi_2\Lambda(y))} \qquad [\lambda(y) \text{ cancels}]$$

$$m'(a) = -(\log \mathcal{L})''(a) = -\operatorname{Var}(V \mid \text{survived to } a) \qquad \text{[Laplace properties]}$$

- V degenerate: m constant, ratio $= \varphi_1/\varphi_2$ for all y.
- $\operatorname{Var} V > 0$: m strictly decreasing. For $y > 0$, $\varphi_1\Lambda(y) > \varphi_2\Lambda(y) \Rightarrow m(\varphi_1\Lambda) < m(\varphi_2\Lambda) \Rightarrow$ ratio $< \varphi_1/\varphi_2$.
- $y = 0$: $\Lambda(0) = 0$, second factor $= m(0)/m(0) = 1$.
- The ratio can even fall below 1 (hazards cross); only the drift below φ_1/φ_2 is used.

Back

Proof: MPH identification in three steps

Step 1. With $\varphi(x^*) = 1$:

$$\frac{\bar{\theta}(y \mid x)}{\bar{\theta}(y \mid x^*)} = \varphi(x)\,\frac{m(\varphi(x)\Lambda(y))}{m(\Lambda(y))} \xrightarrow{y \downarrow 0} \varphi(x)\,\frac{m(0)}{m(0)} = \varphi(x) \quad [\Lambda(y) \to 0;\ m(0) = \mathbb{E}V < \infty,\ \text{continuity}]$$

Step 2. With $\Lambda(y^*) = 1$:

$$\bar{S}(y^* \mid x) = \mathcal{L}\big(\varphi(x)\big) \qquad [\bar{S}(y \mid x) = \mathcal{L}(\varphi(x)\Lambda(y))]$$

$\{\varphi(x)\}$ contains an interval $\Rightarrow$ $\mathcal{L}$ known there $\Rightarrow$ (analyticity) on $(0, \infty)$ $\Rightarrow$ (uniqueness) G.
Step 3.

$$\Lambda(y) = \mathcal{L}^{-1}\big(\bar{S}(y \mid x)\big)/\varphi(x), \qquad \lambda = \Lambda' \qquad [\mathcal{L}\ \text{continuous, strictly decreasing}]$$

Back

Proof: competing risks

$$
\begin{aligned}
\Pr(Y > y, S > s \mid x) &= \mathbb{E}\big[\Pr(Y > y, S > s \mid x, V)\big] && \text{[LIE; } V \perp X\text{]} \\
&= \mathbb{E}\big[\Pr(Y > y \mid x, V)\Pr(S > s \mid x, V)\big] && \text{[} Y \perp S \mid (x, V)\text{]} \\
&= \mathbb{E}\big[e^{-\varphi_Y \Lambda_Y(y) V_Y} e^{-\varphi_S \Lambda_S(s) V_S}\big] && \text{[} S = e^{-\Theta} \text{ each]} \\
&= \mathcal{L}\big(\varphi_Y \Lambda_Y(y), \varphi_S \Lambda_S(s)\big) && \text{[definition of } \mathcal{L}\text{]}
\end{aligned}
$$

$\bar{S}_0(t \mid x)$: set $y = s = t$. Cause-specific hazard:

$$
\begin{aligned}
\bar{\theta}_0^Y(t \mid x) &= \lim_{dt \downarrow 0} \frac{\Pr(t \le Y < t + dt,\, S > t \mid x)}{dt\, \bar{S}_0(t \mid x)} && \text{[definition]} \\
&= \frac{-\partial_y \Pr(Y > y, S > t \mid x)\big|_{y=t}}{\bar{S}_0(t \mid x)} && \text{[derivative of a survival]} \\
&= \lambda_Y(t)\varphi_Y(x)\, \frac{-\mathcal{L}_a(\Lambda_Y(t \mid x), \Lambda_S(t \mid x))}{\mathcal{L}(\Lambda_Y(t \mid x), \Lambda_S(t \mid x))} = \lambda_Y(t)\varphi_Y(x)\, \mathbb{E}[V_Y \mid R_t, x] && \text{[chain rule; survivor mean]}
\end{aligned}
$$

Back

Proof: where the selection lives

Weights $w \propto e^{-aV_Y - bV_S}$ (those at risk), $\mathbb{E}_w$ the corresponding expectation.

$$\partial_a \log \mathcal{L} = \mathcal{L}_a/\mathcal{L} = -\mathbb{E}_w[V_Y] \qquad \text{[differentiate under } \mathbb{E}\text{]}$$

$$\partial_b \partial_a \log \mathcal{L} = \frac{\mathcal{L}_{ab}}{\mathcal{L}} - \frac{\mathcal{L}_a \mathcal{L}_b}{\mathcal{L}^2} = \mathbb{E}_w[V_Y V_S] - \mathbb{E}_w[V_Y]\mathbb{E}_w[V_S] \qquad \text{[quotient rule]}$$

$$= \mathrm{Cov}_w(V_Y, V_S) \qquad \text{[definition]}$$

$$\partial_b \log \bar{\theta}_0^Y = \partial_b \log \mathbb{E}_w[V_Y] = \frac{-\mathrm{Cov}_w(V_Y, V_S)}{\mathbb{E}_w[V_Y]} \qquad [\bar{\theta}_0^Y = \lambda_Y \varphi_Y \mathbb{E}_w[V_Y]\text{; same quotient rule}]$$

Direction example, $\varphi_j = e^{x'\beta_j}$, $d = (\beta_{Y2}, -\beta_{Y1})$:

$$d'\beta_Y = \beta_{Y2}\beta_{Y1} - \beta_{Y1}\beta_{Y2} = 0, \qquad d'\beta_S = \beta_{Y2}\beta_{S1} - \beta_{Y1}\beta_{S2} \neq 0 \qquad \text{[iff } (\beta_Y\ \beta_S) \text{ full rank]}$$

Back

Proof: identification of the competing-risks model (Prop. 2)

Step 1. $\dfrac{\bar\theta_0^Y(t \mid x)}{\bar\theta_0^Y(t \mid x^*)} = \varphi_Y(x)\dfrac{\mathbb{E}[V_Y \mid R_t, x]}{\mathbb{E}[V_Y \mid R_t, x^*]} \xrightarrow{t\downarrow 0} \varphi_Y(x)$ [$R_t \to$ everyone; $V \perp X$; $\mathbb{E}V_Y < \infty$]. Same for φ_S.

Step 2. $\bar S_0(t^* \mid x) = \mathcal{L}(\varphi_Y(x), \varphi_S(x))$. Assn. 3: an open set $\Rightarrow$ $\mathcal{L}$ on $(0,\infty)^2 \Rightarrow G$.

Step 3. Suppose $(a,b) \neq (a',b')$ both satisfy $\bar S_0(t \mid x) = \mathcal{L}(\varphi_Y(x)a, \varphi_S(x)b)$ for all x.

$$\mathcal{L}(pa, qb) = \mathcal{L}(pa', qb') \text{ on an open set of } (p,q) \qquad \text{[Assn. 3]}$$
$$\mathcal{L}(u,v) = \mathcal{L}(ku, mv) \;\; \forall (u,v), \quad k = a'/a, \;\; m = b'/b \qquad \text{[analyticity]}$$

- $k, m \geq 1$ (or both ≤ 1), not both 1: contradicts strict monotonicity of $\mathcal{L}$.
- $k > 1 > m$: $\mathcal{L}(u,v) = \mathcal{L}(k^n u, m^n v) \to \Pr(V_Y = 0)$ (dominated convergence), so $\mathcal{L}$ is constant: contradiction. $k < 1 < m$: symmetric.

Steps are a reconstruction; the paper cites Abbring & Van den Berg (2003, JRSSB).

Back

Proof: no anticipation, data, timing luck

No anticipation $\Theta_{Y^*(s)}(y) = \Theta_{Y^*(s')}(y)$, $y \le \min\{s, s'\}$. Take $s' = \infty$, differentiate in y: $\theta_{Y^*(s)}(y) = \theta_{Y^*(\infty)}(y)$ for $y < s$.

Data $Q_S(y,s) = \Pr(Y > y, S > s, Y > S)$, $Q_Y(y) = \Pr(Y > y, Y < S)$:

$$\bar{S}_0(t) = Q_S(t,t) + Q_Y(t) \qquad \text{[partition by } Y \gtrless S;\ \Pr(Y = S) = 0]$$

$$\bar{\theta}_0^Y(t) = -Q_Y'(t)/\bar{S}_0(t) \qquad \text{[density of untreated exits]}$$

$$\bar{S}_1(y \mid s) = \partial_s Q_S(y,s) \Big/ \lim_{y' \downarrow s} \partial_s Q_S(y',s) \qquad [-\partial_s Q_S\text{: density of } S = s \text{ with } Y > y]$$

Timing luck $E_S := \Theta_S(S \mid x, V) = \varphi_S(x) V_S \Lambda_S(S)$:

$$\Pr(E_S > e \mid x, V) = \Pr\left(S > \Theta_S^{-1}(e) \mid x, V\right) = e^{-e} \qquad [\Theta_S \text{ increasing; } S = e^{-\Theta}]$$

$$\implies S = \Lambda_S^{-1}\big(E_S/(\varphi_S(x)V_S)\big), \quad E_S \sim \text{Exp}(1) \perp (x, V) \qquad \text{[invert]}$$

Back

Proof: the selection factor

$$dG_{R_s}(v) = \frac{\Pr(R_s \mid x, v)\, dG(v)}{\Pr(R_s \mid x)} \propto e^{-a_s v_Y - b_s v_S}\, dG(v) \qquad \text{[Bayes; no anticipation; Assn. 2]}$$

$$dG_s(v) \propto \Pr(s \le S < s + ds \mid R_s, x, v)\, dG_{R_s}(v) \propto v_S\, dG_{R_s}(v) \qquad \text{[Bayes; } \theta_S \propto v_S\text{]}$$

Exit hazards at s: just treated $\lambda_Y \varphi_Y \delta(s \mid s, x) \mathbb{E}_{G_s}[V_Y]$; not yet treated $\lambda_Y \varphi_Y \mathbb{E}_{R_s}[V_Y]$.

$$\mathbb{E}_{G_s}[V_Y] = \frac{\mathbb{E}_{R_s}[V_Y V_S]}{\mathbb{E}_{R_s}[V_S]} = \mathbb{E}_{R_s}[V_Y] + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s}[V_S]} \qquad \text{[reweight by } v_S\text{; } \mathbb{E}XY = \mathbb{E}X\mathbb{E}Y + \mathrm{Cov}\text{]}$$

$$\frac{\text{just treated}}{\text{not yet treated}} = \delta(s \mid s, x)\Big(1 + \frac{\mathrm{Cov}_{R_s}}{\mathbb{E}_{R_s} V_Y\, \mathbb{E}_{R_s} V_S}\Big) \qquad \text{[divide]}$$

$$= \delta(s \mid s, x)\, \frac{\mathcal{L}\, \mathcal{L}_{ab}}{\mathcal{L}_a\, \mathcal{L}_b}\Big|_{(\Lambda_Y(s|x), \Lambda_S(s|x))} \qquad \text{[}\mathbb{E}V_Y = -\tfrac{\mathcal{L}_a}{\mathcal{L}}\text{, } \mathbb{E}V_S = -\tfrac{\mathcal{L}_b}{\mathcal{L}}\text{, } \mathbb{E}V_Y V_S = \tfrac{\mathcal{L}_{ab}}{\mathcal{L}}\text{]}$$

Back

Proof: after the moment of treatment (eq. 10)

$\Theta_Y(y \mid s, x, V) = \varphi_Y V_Y[\Lambda_Y(s) + D(y \mid s, x)]$ for $y > s$ [split the integral at s]. Density g of "treated at s, still unemployed at y":

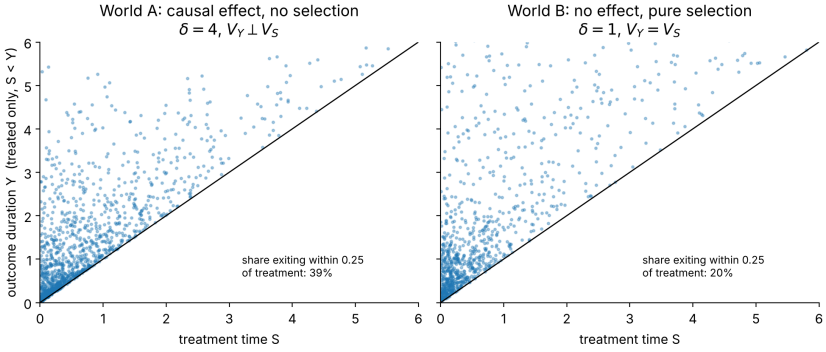
$$
\begin{aligned}
g(y, s \mid x) &= \mathbb{E}\big[f_S(s \mid x, V)\ \Pr(Y > y \mid S = s, x, V)\big] && \text{[LIE; Assn. 2]} \\
&= \mathbb{E}\big[\lambda_S \varphi_S V_S e^{-b_s V_S} \cdot e^{-\varphi_Y V_Y[\Lambda_Y(s)+D]}\big] && [f = \theta e^{-\Theta};\ S = e^{-\Theta}] \\
&= -\lambda_S(s)\varphi_S(x)\, \mathcal{L}_b\big(\Lambda_Y(s \mid x) + \varphi_Y(x) D,\ \Lambda_S(s \mid x)\big) && [\mathcal{L}_b = -\mathbb{E}[V_S e^{-aV_Y - bV_S}]] \\
\bar{S}_1(y \mid s, x) &= \frac{g(y, s \mid x)}{g(s, s \mid x)} = \frac{\mathcal{L}_b(\Lambda_Y(s \mid x) + \varphi_Y D,\ \Lambda_S(s \mid x))}{\mathcal{L}_b(\Lambda_Y(s \mid x), \Lambda_S(s \mid x))} && [D(s \mid s, x) = 0]
\end{aligned}
$$

- $\partial_D \mathcal{L}_b(\Lambda_Y(s \mid x) + \varphi_Y D, \Lambda_S(s \mid x)) = \varphi_Y \mathbb{E}[V_Y V_S e^{\cdots}] > 0$ and $\mathcal{L}_b < 0$: $\bar{S}_1$ strictly decreasing in D $\Rightarrow$ invert.
- $\delta(y \mid s, x) = \partial_y D(y \mid s, x)/\lambda_Y(y)$ [fundamental theorem of calculus].

Local vs. global picture

Back

Backup: local vs. global dependence



Left: causal effect, no selection; exits bunch right after treatment at every S. Right: no effect, pure selection; S and Y correlated overall.

Back